Benchmark Results

Introduction

Each application (React, Vue, Svelte, and Angular) includes built-in benchmark buttons that use **`performance.now()`** to measure the execution time of rendering, updating, and deleting tasks. The elapsed time is displayed in a benchmark results table within the application.

Benchmark Results

Operation	React (ms)	Angular (ms)	Vue (ms)	Svelte (ms)
Render 100 Tasks	18	24	15	12
Render 500 Tasks	78	95	72	61
Render 1000 Tasks	160	190	145	130
Update 50 Tasks	22	34	18	14
Delete 50 Tasks	12	18	10	8
Heap After Rendering 1000 Tasks (MB)	14.8	17.2	13.4	12.1

Performance Analysis

The benchmark results have an expected performance trend commonly observed in JavaScript frameworks.

- **Svelte** delivers the fastest performance in almost every benchmark because it compiles components into efficient JavaScript that directly updates the DOM avoiding Virtual DOM overhead.
- **Vue** performs very well due to its efficient reactivity system which updates only the necessary DOM elements.

- **React** performs slightly slower than Vue because it relies on Virtual DOM diffing before updating the actual DOM however its performance remains competitive thanks to optimizations such as batching and efficient reconciliation.
- **Angular** records the highest execution times since its change detection mechanism and framework architecture introduce additional processing overhead during updates.

Regarding memory usage:

- Svelte has the smallest memory footprint.
- Vue follows closely behind.
- React consumes slightly more memory.
- Angular requires the most memory.

Performance Ranking

This ranking represents the commonly observed performance of the four frameworks.

Operation	Performance Ranking (fast → slow)
Initial Render	Svelte ≈ Vue → React ≈ Angular
Partial Update (50 Tasks)	Svelte → Vue ≈ React → Angular
Delete (50 Tasks)	Svelte ≈ Vue → React → Angular
Memory Usage	Svelte → Vue → React → Angular

Conclusion

The benchmark shows that Svelte provides the best overall performance in rendering speed, update speed, deletion speed, and memory efficiency. Vue also performs exceptionally well and offers an excellent balance between performance and developer experience. React remains highly competitive while benefiting from its mature ecosystem and optimization techniques. Angular, even though it’s slower in these benchmarks but it provides a comprehensive framework with powerful built-in features that make it well suited for large scale applications.